

Elyssa Hofgard

ehofgard.github.io

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Massachusetts Institute of Technology, MA

EDUCATION

Massachusetts Institute of Technology Cambridge, MA
PhD Electrical Engineering and Computer Science (EECS)
Sept 2022-present

GPA: 4.80/5.00

Stanford University Stanford, CA
M.S. Computational and Mathematical Engineering
Jan. 2021-June 2022

GPA: 3.97/4.00

Stanford University Stanford, CA
B.S. Physics with Honors, Minors: Mathematics and History
Sept. 2017-June 2021

GPA: 3.91/4.00

RESEARCH & EMPLOYMENT EXPERIENCE

Massachusetts Institute of Technology **Sept 2022 - present**
Graduate Research Assistant, Atomic Architects, Professor Tess Smidt

- Developing symmetry equivariant neural networks for applications in the natural sciences

Lawrence Berkeley National Laboratory **May 2024-present**
Graduate Intern, Dr. Sinéad Griffin & Dr. Dmitriy Morozov

- Characterizing length scales of order in amorphous materials, modeling multipolar order parameters in magnetic materials

Tompkins Laboratory **Sept 2021 - June 2022**
Research Assistant, IRIS-HEP

- Funded as a research assistant through Institute for Research and Innovation in Software for High Energy Physics (IRIS-HEP)
- Project focused on improving Kalman Filter algorithms used in particle track reconstruction

Lawrence Livermore National Laboratory **June 2021 - Sept 2021**
Intern, Data Science Summer Institute (DSSI)

- Project focused on improving seasonal predictions of precipitation and temperature over the western US using deep learning

ATLAS Experiment, CERN **June 2020 - June 2021**
Research Assistant, Professor Lauren Tompkins

- 1 of 11 students selected nationwide for the International Research Experience for Students (IRES) at CERN funded through the National Science Foundation
- Honors thesis explored the use of machine learning techniques to search for dark photons in ATLAS data, improved analysis sensitivity 10 times
- Thesis awarded Firestone Medal for Excellence in Undergraduate Research

Molecular Imaging Instrumentation Laboratory **Jan 2019 - Jan 2021**
Stanford, CA
Research Assistant, Professor Craig Levin

- Led an experiment to develop novel algorithms for analyzing dual-isotope positron emission tomography (PET) systems

Regeneron Genetics Center Jun 2019 - Aug 2019
Tarrytown, NY
Summer Intern, R&D Data Engineering Group

- Led the development of a statistical method to distinguish gene level effects across phenotypes

LIGO Group Jun 2018 - Jan 2019
Stanford, CA
Research Assistant, Dr. Brian Lantz

- Designed a wind fence for seismic isolation at the LIGO Hanford Site

HONORS & AWARDS **QuARC, MIT QSEC Annual Conference** Jan 2024
Best Poster in Quantum Algorithms and Machine Learning Session.

Department of Energy Computational Science Graduate Fellowship 2023
Accepted, up to 4 years of funding

National Defense Science and Engineering Graduate Fellowship 2023
Declined for DOE CSGF

MIT Advanced Television and Signal Processing Fellowship Sept 2022

NSF GRFP Honorable Mention April 2022

Firestone Medal for Excellence in Undergraduate Research August 2021

- Award given to the top ten percent of all honors theses in the social sciences, natural sciences, and engineering and applied sciences

American Physical Society Far West Section Meeting Nov 2019
Honorable Mention for Best Undergraduate Poster

PUBLICATIONS **Hofgard, Elyssa**, Kyuchool Min, Nofit Segal, David W. Mittan-Moreau, Aria Mansouri Tehrani, Vanessa Oklejas, Jigyasa Nigam, Daniel W. Paley, Aaron S. Brewster, Tess Smidt. *Learning Lattice Parameters from Powder X-Ray Diffraction Data Using Invariants*, 2026, <https://arxiv.org/abs/2607.21829>. Under review.

Lawrence, Hannah*, **Elyssa Hofgard***, Vasco Portilheiro, Yuxuan Chen, Tess Smidt, Robin Walters, *To Augment or Not to Augment? Diagnosing Distributional Symmetry Breaking*, 2026, *International Conference on Learning Representations*, <https://arxiv.org/abs/2510.01349>. *denotes equal contribution.

Artificial intelligence for science in quantum, atomistic, and continuum systems, Foundations and Trends in Machine Learning, 2025, Vol. 18: No. 4, pp 385-912, <https://doi.org/10.1561/22000000115>.

Hofgard, Elyssa, R. Wang, R. Walters, and T. Smidt. "Relaxed Equivariant Graph Neural Networks." *Geometry-grounded Representation Learning and Generative Modeling Workshop (GRaM) at Forty-first International Conference on Machine Learning*, July 2024. <https://arxiv.org/abs/2407.20471>.

Wang, R., **Elyssa Hofgard**, H. Gao, R. Walters, and T. Smidt. "Discovering Symmetry Breaking in Physical Systems with Relaxed Group Convolution." *Forty-first International Conference on Machine Learning*, February 2024. <https://arxiv.org/abs/2310.02299>.

Cotter, D., **Elyssa Hofgard**, A. Szpiech, and N. Rosenberg. "A Rarefaction Approach for Measuring Population Differences in Rare and Common Variation." *Genetics*, April 2023.

The ATLAS collaboration. “Search for dark photons from Higgs boson decays via ZH production with a photon plus missing transverse momentum signature from pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector.” *J. High Energ. Phys.* 2023, 133. [https://doi.org/10.1007/JHEP07\(2023\)133](https://doi.org/10.1007/JHEP07(2023)133). *Work recognized by the 2025 Breakthrough Prize in Fundamental Physics.*

Garg, R.B., **Elyssa Hofgard**, L. Tompkins, and H. Gray. “Exploration of Different Parameter Optimization Algorithms within the Context of ACTS Software Framework.” 2022. <https://doi.org/10.48550/arXiv.2211.00764>.

Hofgard, Elyssa, G. Chinn, and C. Levin. “Simultaneous Multi-Isotope PET: A Computational Framework for Line of Response (LOR) Identification.” 2020 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC), 2020. <https://doi.org/10.1109/NSS/MIC42677.2020.9507891>.

LIGO Scientific and Virgo Collaborations. “Open Data from the First and Second Observing Runs of Advanced LIGO and Advanced Virgo.” 2019. <https://arxiv.org/abs/1912.11716>.

CONFERENCES & TALKS

Hofgard, Elyssa, Guy Moore, Omar Ashour, Ella Banyas, Tess Smidt, Sinead M. Griffin. *Using Equivariant Neural Networks to Learn Multipolar Order Parameters*. Oral Presentation at APS Global Physics Summit 2026.

Hofgard, Elyssa, Dmitriy Morozov, Musa A Hussien, Temuujiin Bayaraa, Sinead M Griffin, Tess E Smidt. *Applications of Topological Data Analysis (TDA) for Studying Structural Motifs in Amorphous Bi_2Se_3* . Oral Presentation at APS Global Physics Summit 2025.

Using $E(3)$ -Equivariant Neural Networks (ENNs) to Uncover Symmetry-Implied Missing Information. Computational Research in Boston and Beyond (CRIBB) Seminar, March 2025.

Applications of Euclidean Neural Networks for the Design and Understanding of Physical Systems. Plenary talk at The 5th International Conference on Data-Driven Plasma Science, Berkeley, CA. Given on behalf of Tess Smidt.

Hofgard, Elyssa, A.M. Tehrani, and T. Smidt. *Learning Crystal Structure from Powder X-Ray Diffraction Data Using Invariants*. Oral Presentation at 2024 American Physical Society March Meeting.

Hofgard, Elyssa, R. Wang, and T. Smidt. *Using Symmetry-Equivariant Neural Networks (ENNs) to Uncover Symmetry-Implied Missing Information*. Oral Presentation at 2023 Materials Research Society Fall Meeting.

Hofgard, Elyssa et al. *Exploration of Machine Learning (ML) Methods for Determining Unit Cell Parameters from Powder XRD Patterns*. Oral Presentation at 2023 Materials Research Society Fall Meeting.

Hofgard, Elyssa, and L. Tompkins. *Cuts Optimization and Machine Learning Models for Dark Photon Signal-Background Discrimination with the ATLAS Detector*. Oral presentation at 2021 APS April Meeting.

Hofgard, Elyssa. *An Algorithmic Approach to Identify Non-Pure Positron Emitters for Positron Emission Tomography (PET) Imaging Applications*. Poster presented at 2019 Annual Meeting of the APS Far West Section.

PROFESSIONAL SERVICE

- Reviewer, International Conference on Learning Representations (ICLR), 2026
- Reviewer, Conference on Neural Information Processing Systems (NeurIPS), 2026

- Reviewer, Geometry-grounded Representation Learning and Generative Modeling (GRaM) Workshop at ICML/ICLR, 2024, 2026
- Reviewer, Machine Learning and the Physical Sciences (ML4PS) Workshop at NeurIPS, 2024-2025

TEACHING & OUTREACH

- Mentor, MIT Undergraduate Research Opportunities Program, 2024-present
 - Supervised an undergraduate researcher (Kyuchool Min) on representation learning for XRD data, resulting in co-authorship on Hofgard et al. (2026)
- Co-President of Graduate Women in Course 6 (GW6), 2023-2024
 - Elected as co-president of GW6, an organization to support graduate women in EECS at MIT
- MIT EECS Graduate Student Council (GSC) Representative, 2022-2023
- Teaching & Mentoring, Physics, Stanford University, 2019-2021
 - Teaching Assistant, Physics 21: Mechanics, Fluids, and Heat, Fall 2020
 - Learning Assistant, Physics 41E: Mechanics, Winter 2020
 - Tutor, Schwab Learning Center, provided individual tutoring for students with learning differences; tutored math courses including linear algebra, multivariable calculus, and differential equations
- Leadership, Physics Undergraduate Women and Minorities at Stanford (PUWMAS), 2018-2020
 - Co-President, 2019-2020; Director of Community Development, 2018-2019
- Undergraduate Representative, Physics Undergraduate Studies Committee, 2020-2021
 - Selected as one of two undergraduate representatives to offer ideas on course design and department culture